

Supplementary Information

Intricacies of Human-AI Interaction in Dynamic Decision-Making for Precision Oncology

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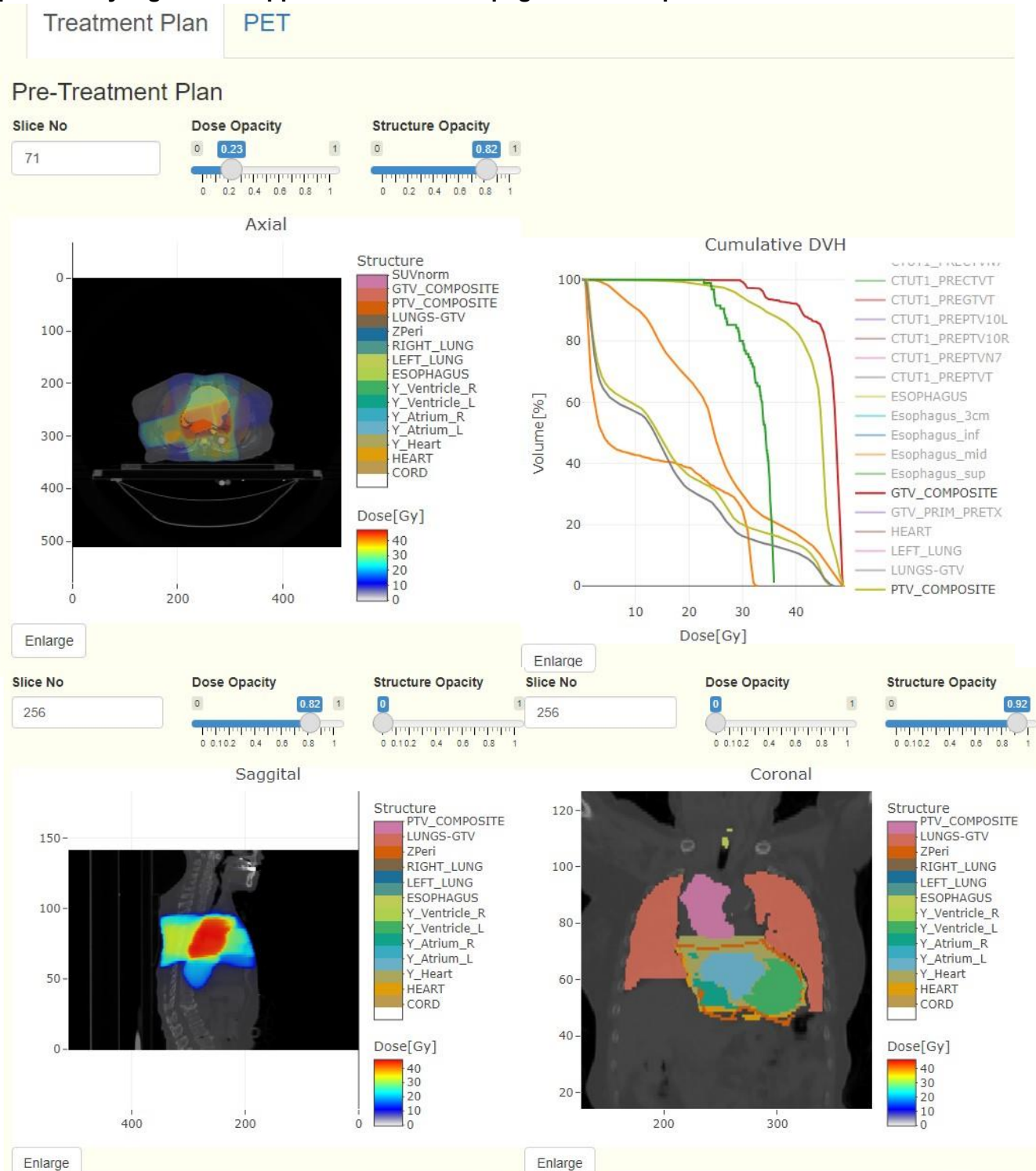
Supplementary Table 1: Patient's Clinical Information

NSCLC											
Pat No	Sex	Age	Cancer Stage	Smoking History	COPD	CVD	Hypertension	Histology	Chemo	KPS	GTV [cc]
1	Female	56-60	1	Yes	No	No	No	Poorly Differentiated	Yes	90	19
2	Male	56-60	3	Yes	No	No	Yes	Adenocarcinoma	Yes	100	207
3	Female	81-85	3	No	No	No	No	Adenocarcinoma	Yes	80	19
4	Female	56-60	3	Yes	No	No	Yes	Adenocarcinoma	Yes	90	58
5	Male	56-60	3	Yes	No	No	Yes	Squamous Cell Carcinoma	Yes	80	359
6	Female	56-60	3	Yes	No	No	No	Adenocarcinoma	Yes	90	102
7	Male	56-60	3	Yes	No	No	No	Squamous Cell Carcinoma	Yes	90	180
8	Female	61-65	3	Yes	Yes	No	No	Squamous Cell Carcinoma	Yes	70	113

HCC													
Pat No	Sex	Age	Cirrhosis	No of Active lesion	PVT	No of Pre-SBRT Systemic Therapies	No of Pre-SBRT Liver-Directed Therapies	Presence of Extrahepatic Disease	No of Prior Liver Occurrences	Previously Treated	ECOG-PS	GTV [cc]	Liver minus GTV [cc]
1	Female	76-80	Yes	1	No	0	2	No	1	Yes	0	17.02	1641
2	Female	76-80	Yes	2	No	0	4	No	3	Yes	0	0.75, 0.92	2546
3	Female	71-75	Yes	4	No	0	2	No	2	Yes	1	4.32, 1.92, 6.32, 0.56	1346
4	Female	76-80	Yes	1	No	0	2	No	2	No	1	2.2	1416
5	Male	56-60	Yes	1	No	0	2	No	3	No	0	9.9	1714
6	Female	56-60	Yes	1	No	0	3	No	1	Yes	1	4.3	2287
7	Female	56-60	Yes	1	Yes	0	0	No	0	No	0	469	1604
8	Female	76-80	Yes	1	No	0	5	No	4	Yes	0	5.51	1281
9	Female	61-65	Yes	1	No	0	1	No	0	No	0	2.04	1533

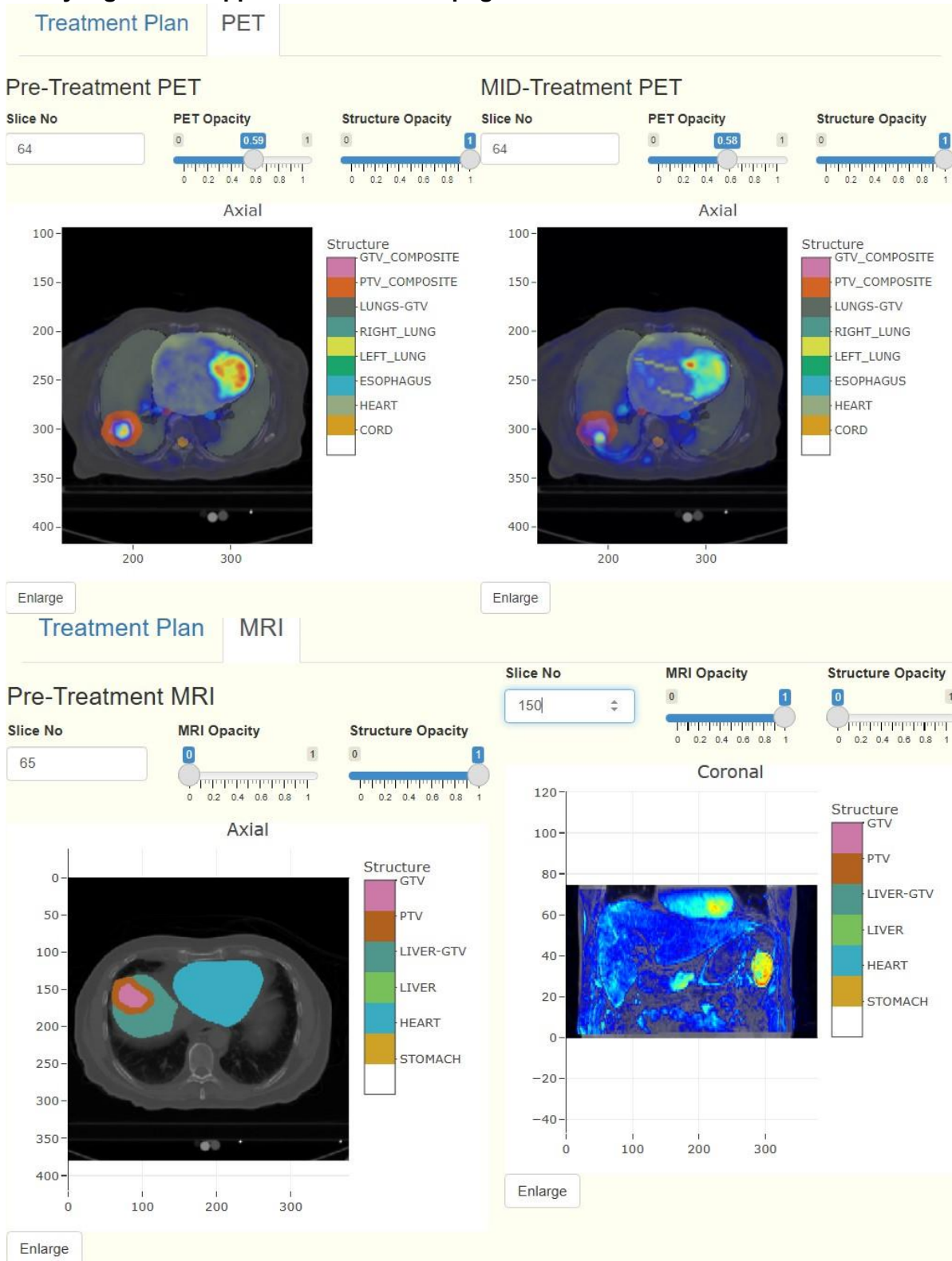
Supplementary Table 1: Patient's clinical information provided to the Evaluators. For NSCLC, the columns names are patient's sex, age (exact age was provided during evaluation), cancer stage, smoking history (binary), chronic obstructive pulmonary disease status (COPD, binary), cardiovascular disease status (CVD, binary), hypertension status (binary), histology (categorical), chemo status (binary), Karnofsky performance status (KPS), and gross tumor volume (GTV, in cc). For HCC, the column names are patient's sex, age, cirrhosis status (binary), number of active lesions, portal vein thrombosis (PVT) status (binary), number of pre-SBRT line of systemic therapies, number of pre-SBRT liver-directed therapy, presence of extrahepatic disease status (binary), number of prior liver occurrence, previous treatment status (binary), eastern cooperative oncology performance status (ECOG-PS), gross tumor volume (GTV, in cc), and Liver minus GTV (in cc).

Supplementary Figure 1: Snippets of Unassisted page treatment plan viewer



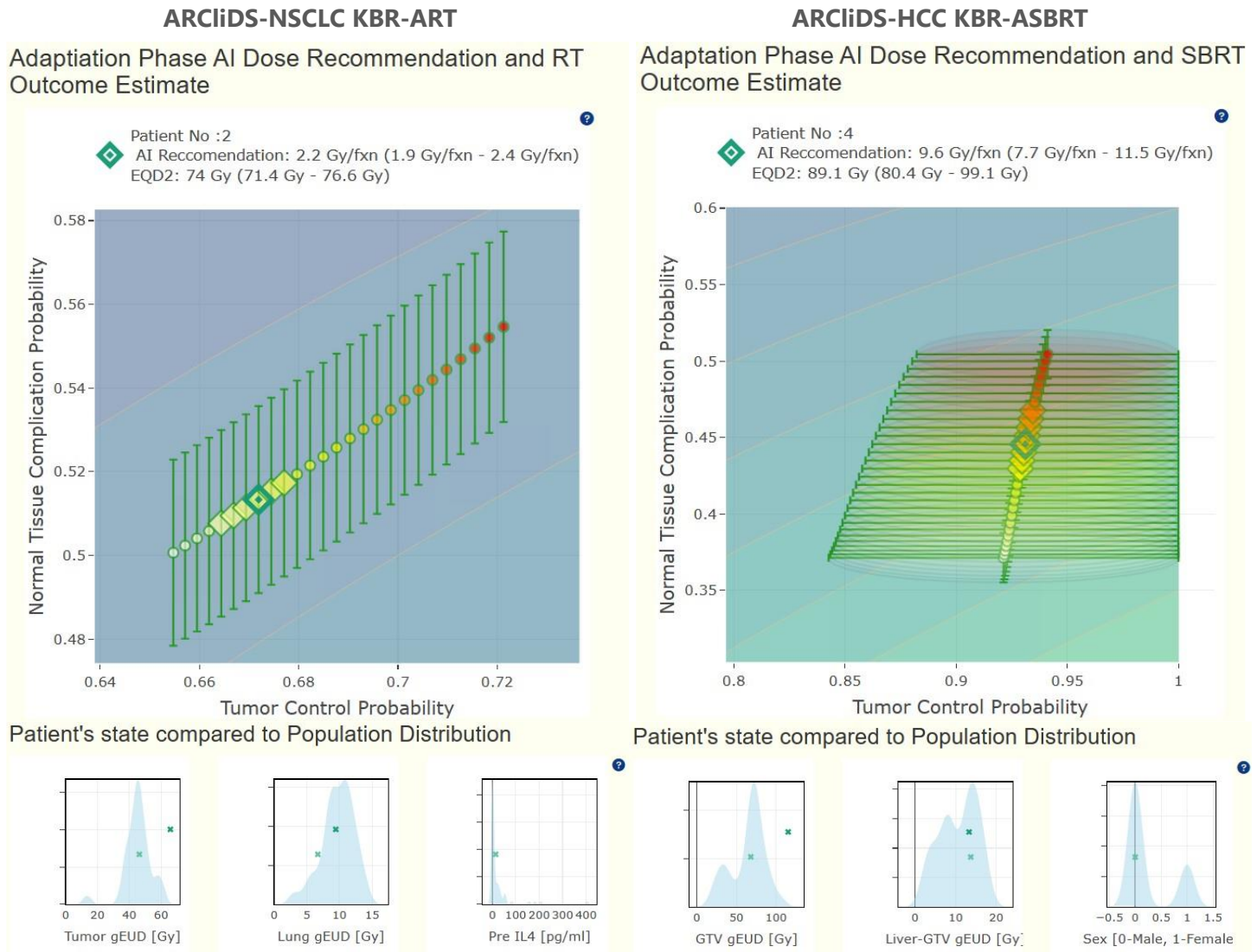
Supplementary Figure 1: Snippets of Unassisted Page treatment plan viewer. The viewer, built on Plotly library, includes Axial, Sagittal, and Coronal view of patients' superimposed CT scan, 3D dose distribution, and Structures, along with their cumulative DVH. Structure and dose values are color coded with discrete and continuous color map, respectively. In addition to the native Plotly controls, the viewer includes three extra controls: slice number input box, dose opacity slider, and structure opacity slider. To demonstrate the controls' functionality, the Axial view has been adjusted to show both dose and structure, the Sagittal to show only dose, and Coronal to show only structure. To demonstrate Plotly's native interactive control, the Coronal View has been zoomed with Plotly's zoom option which has been tweaked to preserve the physical aspect ratio. The native Plotly's option in the cumulative DVH viewer also lets the user select wanted or deselect unwanted histograms. Both NSCLC and HCC Evaluation modules follow the same design.

Supplementary Figure 2. Snippets of Unassisted page PET/MRI viewer



Supplementary Figure 2. Snippets of Unassisted page PET/MRI viewer. The top row shows pre- and mid-treatment PET scans of NSCLC patients in Axial viewer placed in a side-by-side fashion for easy visual comparison. Not shown here are the Sagittal and Coronal Viewers. Bottom row shows pre-treatment MRI scans of HCC patients in Axial and Coronal viewer. Not shown here is the Sagittal Viewer. The viewers superimpose CT scan, PET/MRI image, and structure and include three controls: slice number input box, PET/MRI opacity slider, and structure opacity slider. To demonstrate the controls' functionality, the top row viewers are adjusted to show both PET image and Structure, the Axial MRI viewer is adjusted to show only Structure, and the Coronal MRI viewer to show only MRI scans. The top-row viewers have been zoomed-in using Plotly's native interactive control which has been tweaked to preserve physical aspect ratio.

Supplementary Figure 3. Snippets of AI-assisted Pages showing ARCIIDS recommendation, estimated outcome, and feature distribution panels



Supplementary Figure 3. Snippets of AI-assisted Pages showing ARCIIDS recommendation, estimated outcome, and feature distribution panels. An aggregation of 5 AI-models is included in the Evaluation Module. The AI-recommendation is provided as mean $\pm$ sem daily dose fractionation for the adaptive phase of KBR-ART/KBR-ASBRT treatment paradigm. In addition, corresponding total equivalent dose in 2 Gy fractions (EQD2) is provided. Estimated outcome is provided in the outcome space spanned by tumor control probability (TCP) and normal tissue complication probability (NTCP) for adaptive daily dose fractionation value ranging from 1.5 to 4.0 Gy/frac for NSCLC KBR-ART and from 1.0-15.0 Gy/frac for HCC KBR-ASBRT and color-coded continuously from yellow (lowest dose) to red (highest dose). The outcome corresponding to the AI-recommendation are marked with the green diamond markers and the model uncertainty (sem) is given as error bars in both TCP and NTCP direction. Note that data label class-imbalance can give rise to asymmetry in uncertainty estimate. The background of the outcome space is colored according to the AI reward function, $r = TCP(1 - NTCP)$, which is highest at $(TCP, NTCP) = (1, 0)$. Patient's feature value is presented with population density in the background to provide "whereabouts" about the patient. Not shown here are the feature distribution plots for the remaining 10 features. Help push buttons are included for additional information including feature descriptions. All the viewers are built using Plotly and comes with interactive native controls, such as zooming, selection, etc.

Supplementary Table 2: List of Software and Library

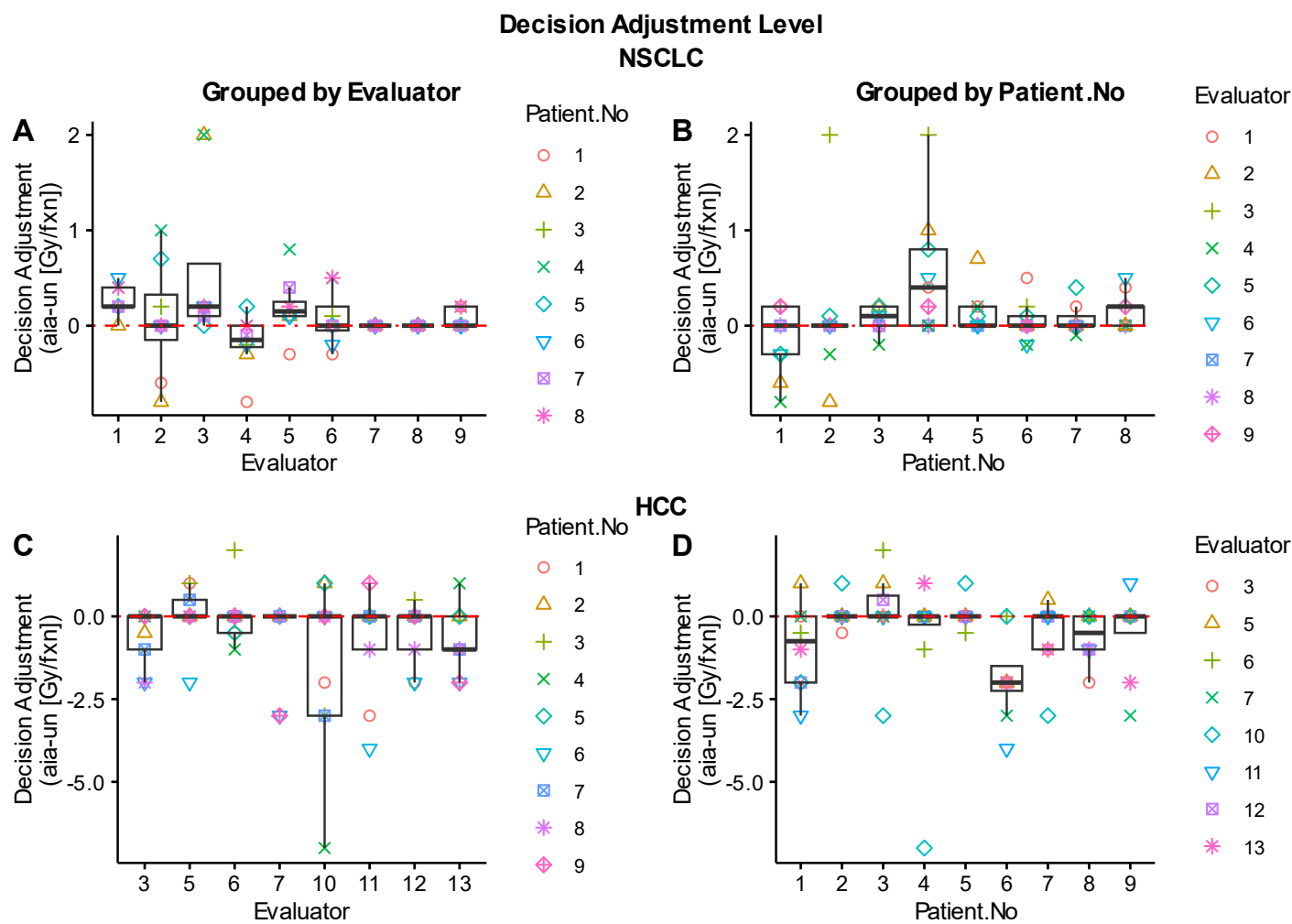
RESOURCE	SOURCE	IDENTIFIER
Python (v3.9.12)	Python Software Foundation	RRID:SCR_008394; https://www.python.org/
NumPy (v1.21.5)	Python package	RRID:SCR_008633; http://www.numpy.org
SciPy (v1.7.3)	Python package	RRID:SCR_008058 http://www.scipy.org/
Pandas (v1.4.2)	Python package	RRID:SCR_018214; https://pandas.pydata.org
PyTorch (v1.11.0)	Python package	RRID:SCR_018536; https://pytorch.org/
PyTorch Geometric (v2.0.4)	Python package	https://pytorch-geometric.com/ ; https://github.com/pyg-team/pytorch_geometric
PyDicom (v2.3.1)	Python package	RRID:SCR_002573; https://pydicom.github.io/
Dicompyler-Core (v0.5.5)	Python package	https://github.com/dicompyler/dicompyler-core
Scikit-Image (v0.19.3)	Python package	RRID:SCR_021142; https://scikit-image.org/
R (v4.2.1)	R Foundation	RRID:SCR_001905; http://www.r-project.org/
Shiny (v1.8.0)	R package	RRID:SCR_001626; http://www.rstudio.com/shiny/
Reticulate (v1.34.0)	R package	https://rstudio.github.io/reticulate/
Plotly (v4.10.3)	R package	RRID:SCR_013991; https://plotly.com/r/
Googlesheets4 (v1.1.1)	R package	https://github.com/tidyverse/googlesheets4
GoolgeDrive (v2.1.1)	R package	https://github.com/tidyverse/googledrive
shinyapps.io (standard subscription)	Posit	https://www.shinyapps.io/ https://docs.posit.co/shinyapps.io/index.html
Power Point (v.2401) [office 365]	Microsoft Crop.	RRID:SCR_023631; https://www.microsoft.com/en-my/microsoft-365/powerpoint

Supplementary Table 3: Evaluators Summary

NSCLC						HCC					
SN	Eval. ID	Exp. (yrs)	Spec.	Inst.	Status	SN	Eval. ID	Exp. (yrs)	Spec.	Inst.	Status
Residents											
1	1	4	Breast	1	Complete	13	3	4	General	1	Complete
2	3	4	General	1	Complete	14	6	2	General	2	Complete
3	4	2	General	2	Complete	15	7	3	General	2	Complete
4	6	2	General	2	Complete	16	13	4	General	1	Complete
5	7	3	General	2	Complete	17	D	3	General	2	Didn't Retake
Resident during training/Physician during Evaluation											
6	2	5	Prostate	1	Complete	18	12	5	GI	1	Complete
Physicians											
7	5	16	CNS	1	Complete	19	5	16	CNS	1	Complete
8	8	21	Prostate	1	Complete	20	10	14	Liver	2	Complete
9	9	17	Lung	1	Complete	21	11	7	GI	1	Complete
10	A	30	GU	1	Didn't retake	22	E	24	GI	2	Didn't retake
11	B	8	Thoracic	1	Didn't Complete	23	F	-	Liver	1	Didn't Take
12	C	-	Lung	1	Didn't Take	24	G	-	GI	1	Didn't Take

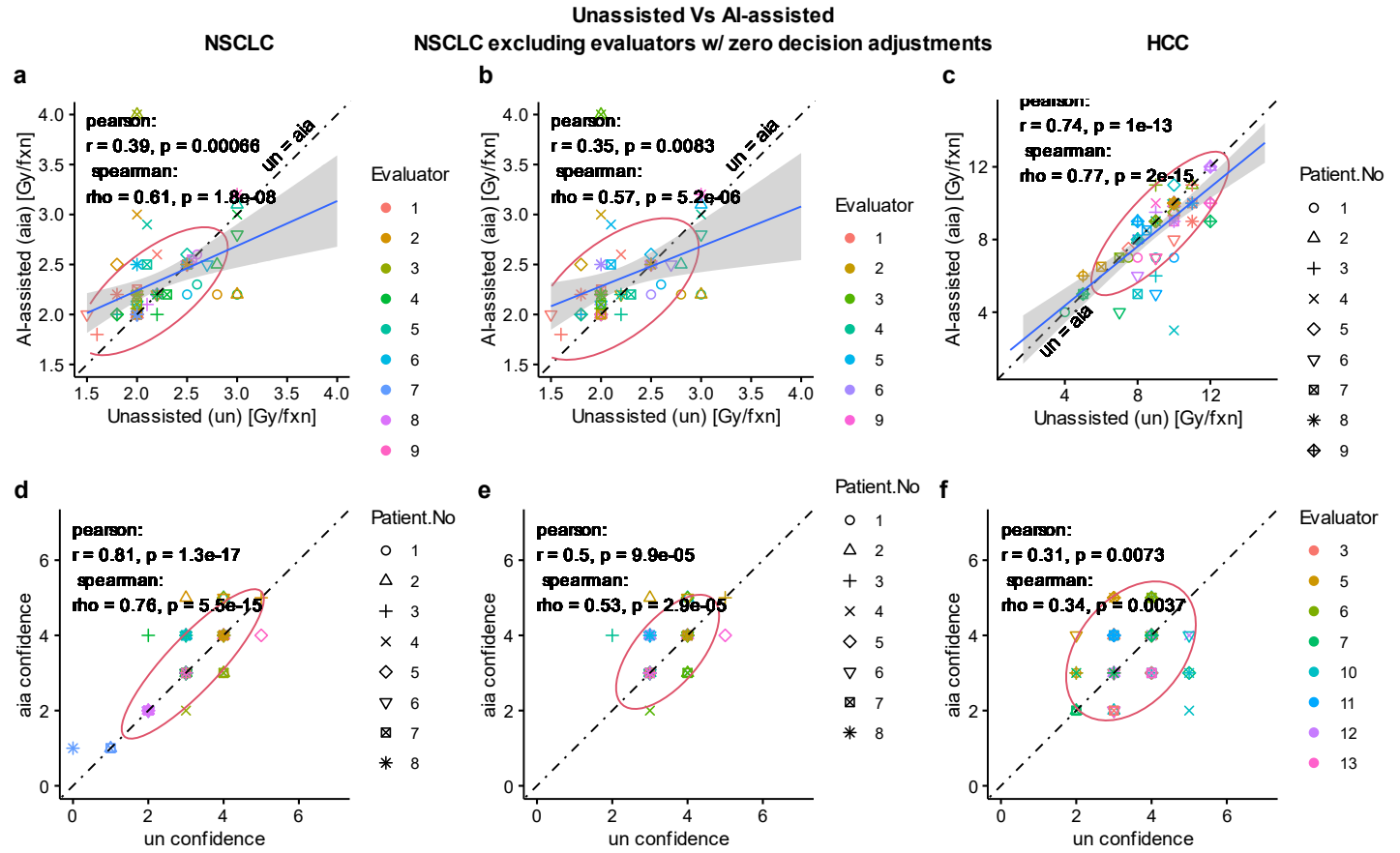
Supplementary Table 3: Summary of volunteer evaluators and their relevant information including those who completed the evaluation, didn't take evaluation after initial training/meeting, and those who took the evaluation initially but declined to retake after discovery of a bug and debugging. The names of evaluators and institutions have been de-identified. Four out of 13 evaluators volunteered to take both evaluations hence 17 evaluations by 13 evaluators. Evaluators who completed the evaluation have been de-identified with numbers and rest with alphabets. Note the special classification for the two evaluators who were resident during the initial training/meeting and went on the pass their board certification and formally accepting physician position before completion of evaluation. **Abbreviation:** **Eval. ID:** Evaluator Identification number, **Exp.:** Experience, **Spec.:** Specialization, **Inst.:** Institution.

Supplementary Figure 4: Decision Adjustment Level



Supplementary Figure 4: Decision Adjustment Level grouped by Evaluators and Patient Number. Box plots **a** and **b** summarize decision adjustment for NSCLC and box plots **c** and **d** for HCC. The plots grouped by evaluators are marked and color coded by patients and vice versa. Box plots include center line: median, box limits: upper and lower quartiles; whiskers: 1.5x interquartile range; and points: outliers

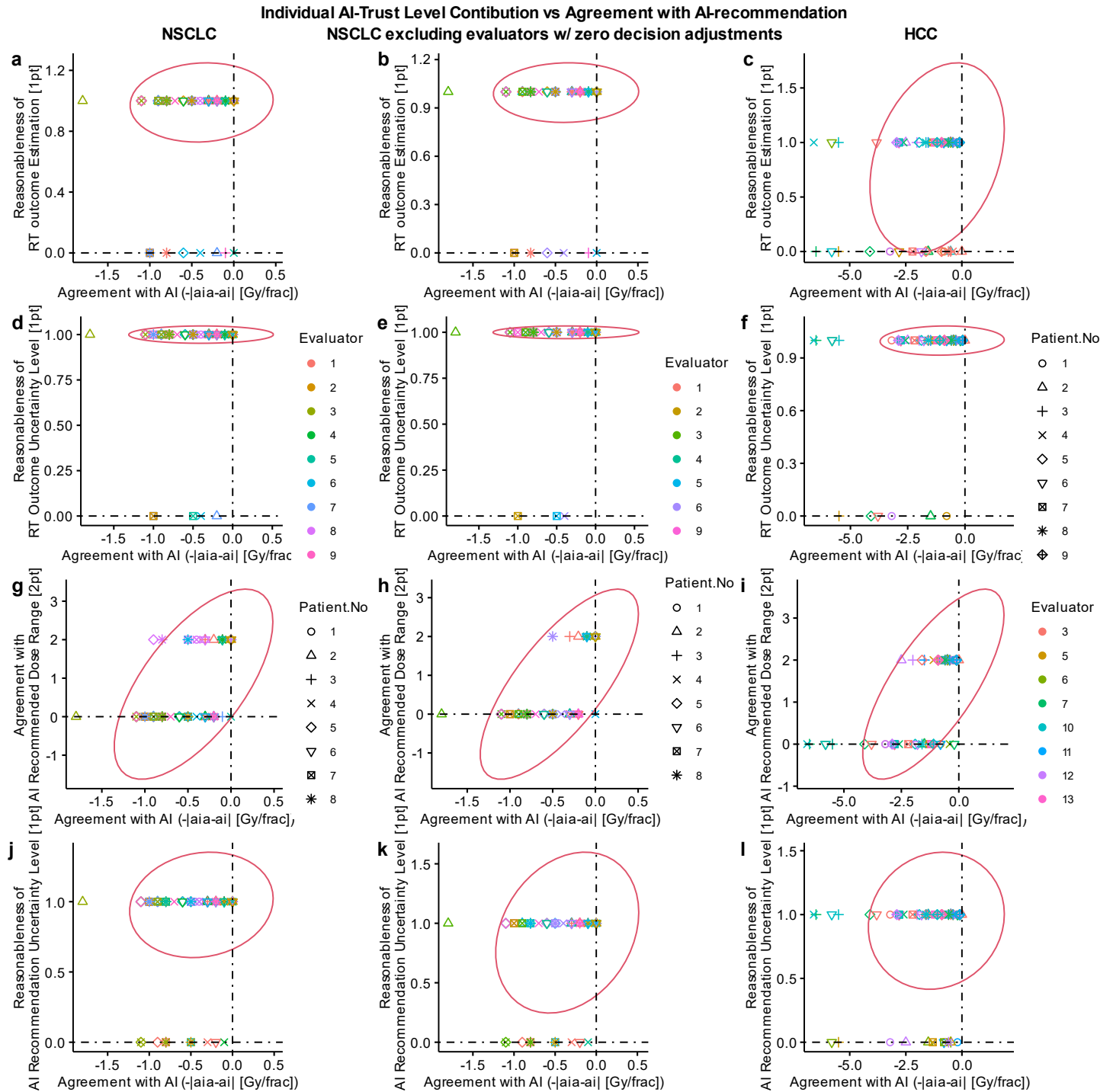
Supplementary Figure 5: Correlation between Unassisted and AI-assisted Decision and Confidence



Supplementary Figure 5: Analysis of decision confidence and confidence level. The first, second, and third column of figures corresponds to NSCLC, NSCLC excluding Evaluators with zero decision adjustment, and HCC, respectively, and all the plots are colored by Evaluators and marked coded Patient Number. Note, all evaluators in HCC adjusted at least one of their decisions. **Scatter plots a, b, and c** show the relationship between unassisted (*un*) and AI-assisted decision (*aia*) and present the Pearson and Spearman correlation coefficient with p-value (two-sided t test), covariance ellipse (95% confidence), a solid linear fit line and uncertainty (95% confidence interval), and the dot-dashed null-hypothesis line $un = aia$ which represents absence of AI-influence. The $ai - un = 0$ line corresponds to complete agreement between unassisted decision and AI-recommendation, whereas $aia - un = 0$ line corresponds to absence of decision change. **2D Scatter plots d, e, and f** show the relationship between evaluators' unassisted decision (*un*) confidence and AI-assisted decision (*aia*) confidence (scaled 0-5, 5 being highest).

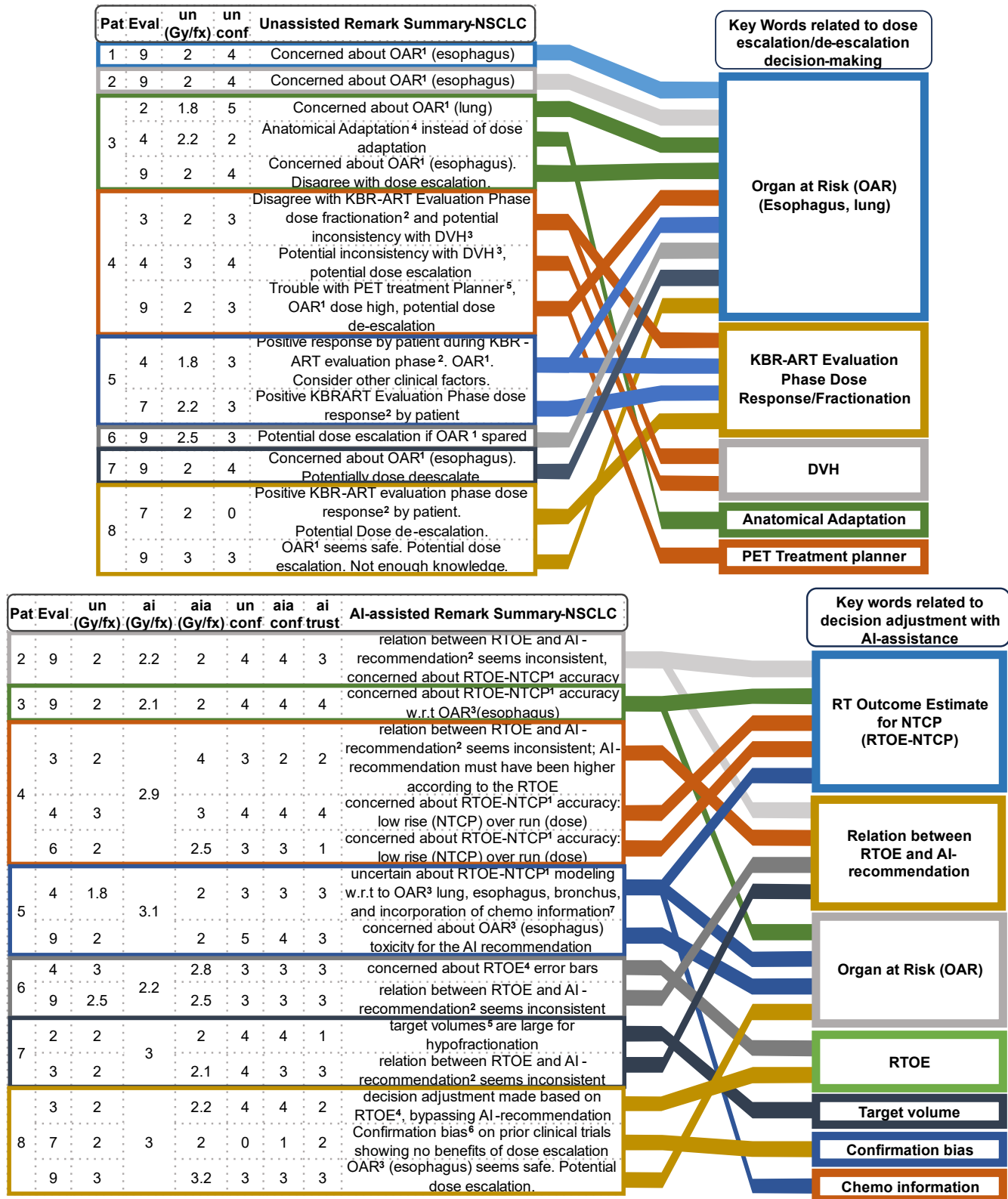
We investigated the correlation between *un* and *aia*. Under null, *un* and *aia* should show perfect (or near perfect) correlation and the tuple (*un*, *aia*) should distribute close to the $un = aia$ line; however, as shown in **Figures a and c**, for NSCLC, we found Pearson correlation coefficient (r) of 0.39 ($p < 0.001$) and spearman correlation coefficient (ρ) of 0.61 ($p < 0.001$), and for HCC, $r = 0.74$ ($p < 0.001$) and $\rho = 0.77$ ($p < 0.001$) which are not near perfect. Pearson coefficient represents level of linear relationship between *un* and *aia* however, unlike Spearman rank correlation, it is very sensitive to extreme value which is seen from the drastic difference between NSCLC r and ρ . In addition, we investigated evaluators who adjusted at least one of their decisions. As expected, we found a decrease in the correlation value as shown in **Figure b**, i.e. $r = 0.35$ ($p = 0.0083$) and $\rho = 0.57$ ($p < 0.001$). The best fit line for both cases did not align with null hypothesis line $un = aia$. The combination of results from hypothesis testing, frequency analysis, and correlation analysis indicated that AI influences decisions on a case-by-case basis. Then, we investigated the relationship between evaluators self-reported Unassisted decision confidence and AI-assisted decision confidence. Under null (no AI-influence), we would expect *aia conf* and *un conf* to show a perfect (or near perfect) correlation and the tuple (*un conf*, *aia conf*) to distribute close to the $un\ conf = aia\ conf$ line. As shown in **Figures d, and f**, we found a positive correlation between *aia conf* and *un conf* decision: for NSCLC, $r = 0.81$ ($p < 0.001$) and $\rho = 0.76$ ($p < 0.001$) and for HCC, $r = 0.31$ ($p = 0.0073$) and $\rho = 0.34$ ($p = 0.0037$). The correlation coefficient is not as strong as we would expect for the null case. Moreover, for NSCLC excluding evaluators with zero decision adjustment, we obtained $r = 0.5$ ($p < 0.001$) and $\rho = 0.53$ ($p < 0.001$), as shown in **Figure e**, which is considerably less than that of the overall NSCLC, consistent with the alternative hypothesis.

Supplementary Figure 6: Individual AI-Trust Level Contribution vs Agreement with AI-recommendation



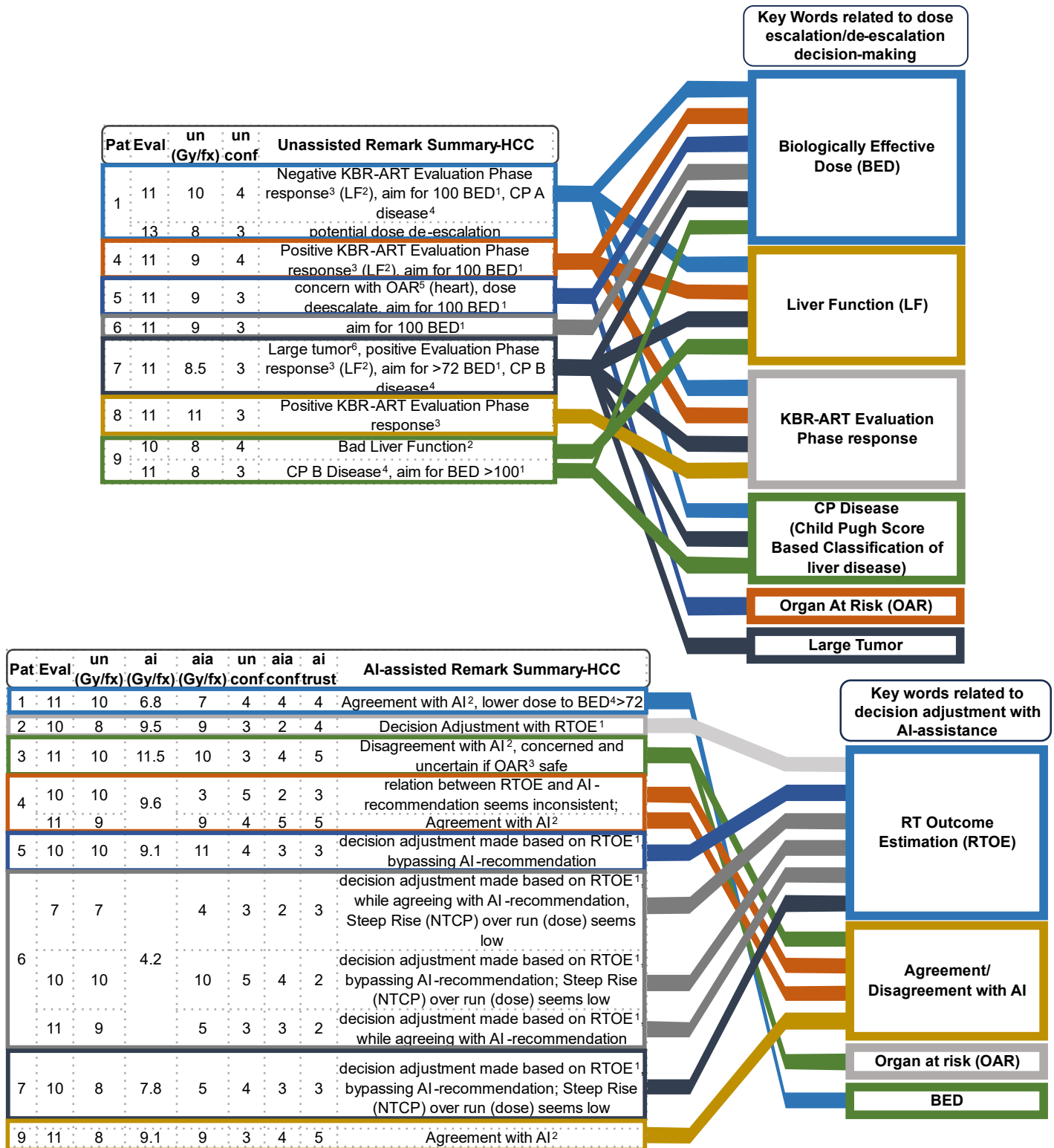
Supplementary Figure 6: Analysis of individual contribution to AI Trust level with respect to agreement between AI-assisted Decision and AI-recommendation. The first, second, and third column of figures corresponds to NSCLC, NSCLC excluding Evaluators with zero decision adjustment, and HCC, respectively, and all the plots are colored by Evaluators and marked coded Patient No. Evaluator's level of trust on AI (scaled 0-5, 5 being the highest) was determined based off of their self-reported agreement with AI's RT outcome estimate (**first two rows**) and AI-recommendation (**last two rows**) which was further divided into average trend (**first and third row**) and uncertainty estimate (**second and last rows**). The individual contributions were Reasonableness of RT outcome Estimation (1 pt), Reasonableness of RT outcome Uncertainty Level (1pt); Agreement with AI recommended Dose range (2pts), and Reasonableness of AI Recommendation Uncertainty Level (1pt). For visual insight, covariance ellipses (95% confidence) are included.

Supplementary Figure 7: Authors summary of Evaluators' Remark for NSCLC.



Supplementary Figure 7: Authors summary of Evaluators' Remark for NSCLC. The tables on the left summarize Evaluators' Remark and present corresponding decision, decision confidence, and AI trust level for Un-assisted and AI-assisted Phase. To present a statistic the repeated keywords are identified and displayed in boxes in the right columns. Following Shankey Diagram's approach, we matched the evaluator's remark to the keyword boxes. The box height is proportional to keyword frequency, and they are sorted from the most frequent to least frequent. Evaluators' full remarks are included in the supplementary material. **Abbreviation-** Pat: Patient No; Eval: Evaluator; un: Unassisted Decision; ai: AI recommendation; aia: AI-assisted Decision; un conf: Unassisted Decision Confidence; aia conf: AI-assisted Decision Confidence; ai trust: AI recommendation trust level

Supplementary Figure 8: Authors summary of Evaluators' Remark for HCC.



Supplementary Figure 8: Authors summary of Evaluators' Remark for HCC. The tables on the left summarize Evaluators' Remark and present corresponding decision, decision confidence, and AI trust level for Un-assisted and AI-assisted Phase. To present a statistic the repeated keywords are identified and displayed in boxes in the right columns. Following Shankey Diagram's approach, we matched the evaluator's remark to the keyword boxes. The box height is proportional to keyword frequency, and they are sorted from the most frequent to least frequent. Evaluators' full remarks are included in the supplementary material. **Abbreviation- Pat:** Patient No; **Eval:** Evaluator; **un:** Unassisted Decision; **ai:** AI recommendation; **aia:** AI-assisted Decision; **un conf:** Unassisted Decision Confidence; **aia conf:** AI-assisted Decision Confidence; **ai trust:** AI recommendation trust level